
Auditing Density Uncertainty in Cryo-EM Beyond a Gaussian Reconstruction Dictionary

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Preliminary experiments; not a completed submission candidate.

Abstract

A confidence interval can be calibrated within a reconstruction dictionary and still miss errors excluded by that dictionary. We investigate this failure in cryogenic electron microscopy (cryo-EM), using conjugate-paired Fourier Gaussians as a computational representation. Our candidate procedure estimates fixed linear density features and audits their bias through the adjoint residual in an explicitly larger voxel space. This separates numerical representation from the density class to which a confidence statement applies. We specialize classical bias-aware inference to the Fourier-slice model, derive explicit bounds for local pose errors, and implement reduced-space and matrix-free solvers with primal-dual diagnostics. Development experiments include repeated bounded-model tests and acquisition geometries from three experimental particle stacks. In fixed-pose tests, dictionary-only intervals can severely undercover an independent map generator; an ambient-space audit restores the assumed-model guarantee. Broader features admit useful intervals, while finer features and restricted viewing angles expose large uncertainty. Density radii, support, Gaussian noise and pose bounds remain explicit assumptions. Comprehensive end-to-end validation, several external baselines and independent review are outstanding; these preliminary results do not establish publication readiness.

1. Introduction

Cryo-EM reconstructs molecular density from noisy projections whose orientations, translations and imaging parameters are imperfectly known. Modern neural and Gaussian

representations offer expressive reconstruction models, but expressiveness alone does not establish which reconstructed features the data support. We focus on a fixed local density average or contrast. This target differs from a conformational population, the uncertainty of an atomic model, or prediction of a noisy held-out particle image.

An uncertainty calculation usually inherits its reconstruction space. If a Gaussian dictionary spans S , a posterior or confidence interval computed in S need not cover density outside S . Improving optimization inside the dictionary does not remove this discrepancy. Independent half reconstructions can also share the same representational restriction. Our working question is whether one can retain Gaussian computation while auditing feature uncertainty in a larger, explicitly declared density class.

We develop three connected components: an adjoint-residual audit that separates fitting and inference spaces; a Fourier-Gaussian treatment of bounded local pose errors, including the density-pose interaction and a uniform nonlinear remainder; and a validation protocol that distinguishes fixed-parameter coverage, Bayesian prior-predictive coverage and experimental-image prediction. We do not claim a new general theory of bias-aware intervals, reduced-space duality or primal-dual optimization. The value of the cryo-EM specialization must be established by its usefulness, assumptions and comparison to existing methods.

2. Related Work and Scientific Scope

Probabilistic reconstruction. Bayesian inference is foundational in cryo-EM (Scheres, 2012). Ullrich et al. (2020) already combine differentiable Fourier-slice reconstruction with variational density uncertainty and discuss the distinction between variance and model bias. Rangan et al. (2024) examine soft modes of the joint volume/alignment likelihood Hessian. Thus neither a probabilistic Fourier representation nor pose-density ambiguity is a new observation. Our candidate emphasizes a finite-sample, bounded-class statement and a check outside the fitting dictionary.

Representations and heterogeneity. CryoDRGN learns a latent-conditioned density family (Zhong et al., 2021a); its latent variation is not automatically calibrated uncertainty in

¹Fourier Cryo Splats research project. Correspondence to: Project repository <<https://github.com/yashpatel5400/fourier-cryo-splats>>.

a homogeneous density. Ab initio neural inference (Zhong et al., 2021b) is a different task from fixed-pose backprojection. Gaussian mixtures and Gaussian splatting are also established (Chen & Ludtke, 2021; Chen & Ling, 2026; Qu et al., 2025; Shekarforoush et al., 2025). We use Fourier kernels as a numerical basis, without treating their existence as the contribution. Continuous-state recovery, conformational populations and pose entanglement are substantial separate problems; the living survey records current RECOVAR, DynaMight, cryoDRGN-AI and CAHRA work with source-level distinctions.

Validation and map confidence. Gold-standard splitting and noise substitution address reconstruction overfitting (Scheres & Chen, 2012; Chen et al., 2013); a shell correlation is not a coverage statement for a local feature. LocScale-2.0 provides calibrated confidence for map enhancement relative to its reference target (Bharadwaj et al., 2026). That target must not be conflated with confidence in an unknown density from raw particles. Map significance relative to background noise, directional resolution, atomic-model reliability and particle-derived confidence likewise answer different questions.

Statistical foundations. Optimal recovery links linear inverse estimation to bias–variance tradeoffs (Donoho, 1994). Honest inference over convex function classes and bias-aware Gaussian critical values are classical (Armstrong & Kolesár, 2018). Our theorems below specialize these principles and identify the assumptions that cryo-EM introduces. The survey and its bibliographic ledger are ongoing; a down-loaded paper is not counted as a completed critical review.

3. Forward Model and Inference Target

With $F(k) = \int \rho(x) e^{-2\pi i k^\top x} dx$, a particle obeys

$$Y_i(q) = C_i(q) e^{-2\pi i q^\top t_i} F(E_i q) + \epsilon_i(q). \quad (1)$$

E_i embeds a detector plane, C_i is its contrast transfer function (CTF), and frequencies and translations use reciprocal units. We realify one representative of each conjugate Fourier pair and whiten the noise. The initial theory conditions on a fixed design and an independent pilot ρ_0 , giving

$$y - \mathcal{A}\rho_0 = \mathcal{A}\delta + Ju + r + \epsilon, \quad \epsilon \sim N(0, I). \quad (2)$$

Here $\delta = \rho - \rho_0$, J is block diagonal over particles, u describes bounded nuisance errors, and r is a bounded remainder. We assume

$$\|\delta\|_H \leq B, \quad \|u_i\| \leq \eta_i, \quad \|r_i\| \leq \gamma_i. \quad (3)$$

H is either a declared Gaussian density subspace or an ambient voxel space with a stated support mask. These

are different classes. A physical density-energy norm includes voxel volume; experiments using an ordinary voxel Euclidean norm explicitly normalize that norm. B , η_i and γ_i are assumptions, not quantities automatically learned from residual fit.

The estimand is $t = \langle \ell, \rho \rangle$, such as a Gaussian-weighted spatial average or the difference of two such averages. The smoothing scale and coordinate frame are specified in physical units. For weights fixed without the inference noise, use

$$\hat{t} = \langle \ell, \rho_0 \rangle + w^\top (y - \mathcal{A}\rho_0). \quad (4)$$

The pilot, dictionary, whitening, target and weights may depend on independent development data. Poses estimated from the inference images violate literal fixed-design independence; supplied consensus poses therefore support only a conditional experimental analysis here.

4. Bias-Aware Confidence and Ambient Auditing

4.1. A conditional certificate

Define $s = \|w\|$ and

$$b_H(w) = B \|\ell - \mathcal{A}^* w\|_H + \sum_i \eta_i \|J_i^\top w_i\| + \sum_i \gamma_i \|w_i\|. \quad (5)$$

Let $q(s, b, \alpha)$ solve

$$\Phi((q - b)/s) - \Phi((-q - b)/s) = 1 - \alpha \quad (6)$$

when $s > 0$, and set $q(0, b, \alpha) = b$.

Theorem 1 (Conditional coverage). *Under Equations (2)–(3), the interval $\hat{t} \pm q(s, b_H(w), \alpha)$ covers t with probability at least $1 - \alpha$, uniformly over the stated class, conditional on the independent design choices.*

Proof. The deterministic estimation error is $\langle \mathcal{A}^* w - \ell, \delta \rangle + \sum_i w_i^\top J_i u_i + \sum_i w_i^\top r_i$. Cauchy–Schwarz bounds its absolute value by b_H . The remaining error is Gaussian with standard deviation s . Coverage of a symmetric interval decreases with the absolute Gaussian mean, so the minimum occurs at mean $\pm b_H$ and equals Equation (6). $\square$

The result is a direct bias-aware inference construction. It is useful only if its assumptions are defensible and its width improves on the no-data interval $\langle \ell, \rho_0 \rangle \pm B \|\ell\|$.

4.2. Why auditing the dictionary matters

Let $S : \mathbb{R}^p \rightarrow H$ have orthonormal columns. A restricted calculation replaces the density term by $B \|S^*(\ell - \mathcal{A}^* w)\|$. Write $h = \ell - \mathcal{A}^* w$. Then

$$\|h\|_H^2 = \|S^* h\|^2 + \|(I - SS^*)h\|_H^2. \quad (7)$$

The second term measures uncertainty discarded by the dictionary. It can be large even when the restricted residual is zero. Evaluating the full adjoint residual repairs this omission for H without estimating a representation-error direction from noisy data. It does not cover density outside the declared ambient grid or support.

Centering in Equation (4) also uses the ambient operator. If a surrogate centers the images instead, the discrepancy $w^\top(\mathcal{A} - \tilde{\mathcal{A}})\rho_0$ requires an additional bound. We do not silently substitute a surrogate forward model into the coverage theorem.

4.3. Optimization and verifiable stopping

We optimize the conservative convex objective

$$\mathcal{F}_H(w) = z_{1-\alpha/2} \|w\| + b_H(w), \quad (8)$$

then report the sharper width from Equation (6). Its dual is

$$\begin{aligned} \max_{v, a, d} \quad & \langle \ell, v \rangle \\ \text{s.t.} \quad & \|v\| \leq B, \quad \|a_i\| \leq \eta_i, \quad \|d_i\| \leq \gamma_i, \\ & \|\mathcal{A}v - Ja - d\| \leq z_{1-\alpha/2}. \end{aligned} \quad (9)$$

Representing each norm by a Euclidean support function proves this dual; continuity and coercivity yield strong duality. A feasible dual iterate supplies a lower bound before convergence. The gap concerns Equation (8), not optimality of the folded-normal width.

For a reduced density space S , a feasible dual vector lifts to Sw in H . Thus a restricted dual lower bound and an ambient primal upper bound bracket the same full-space optimum. We enrich S with the component of $\ell - \mathcal{A}^*w$ orthogonal to it, using only design and target information. Restricted exact optima increase under enrichment; individual ambient primal values need not decrease, so we keep the best feasible iterate. No convergence-rate claim is made for this greedy construction.

Quadratic majorization gives coefficient-space conjugate-gradient solves with particle-block inverses. A classical Chambolle–Pock implementation is an alternative (Chambolle & Pock, 2011). For fixed poses, a full-space matrix-free comparator solves $(\mathcal{A}^*\mathcal{A} + \lambda I)v = \ell$ within majorization. It can outperform enrichment and is included to prevent an unsupported Gaussian speed claim.

5. Fourier Gaussians and Local Pose Errors

We use smooth conjugate pairs

$$\begin{aligned} F_c(k) &= \sum_j c_j g_j(k) + \bar{c}_j g_j(-k), \\ g_j(k) &= e^{-\|k - \mu_j\|^2 / (2\sigma_j^2)}. \end{aligned} \quad (10)$$

This enforces Hermitian symmetry. Hard cutoff kernels from the earlier reconstruction prototype are unsuitable for an unqualified Taylor bound; the uncertainty implementation uses full Gaussians. Their analytic whole-space Gram matrix provides density-energy coordinates $\delta = T\beta$ with $T^*GT = I$, so $\|\beta\| \leq B$ is a density-energy assumption rather than an arbitrary coefficient scale.

Parameterize local rotations by a Lie-algebra vector and translations by detector shifts. Normalize their joint vector so $\|u_i\| \leq 1$, with physical radii incorporated in the operator. With $D_{ia} = \partial A_i / \partial u_{ia}$ at the pilot pose,

$$\begin{aligned} A_i(u_i)(c_0 + \delta) &= A_i(0)(c_0 + \delta) + J_i u_i \\ &\quad + \sum_a u_{ia} D_{ia} \delta + e_i. \end{aligned} \quad (11)$$

If H_i uniformly bounds the pilot’s second derivative along all allowed segments and K_i bounds the corresponding operator derivative, then $\|e_i\| \leq (H_i + BK_i)/2$. Define $T_i(w_i) = [D_{i1}^\top w_i, \dots, D_{iq}^\top w_i]$. The interaction contributes at most

$$B \sum_i \|T_i(w_i)\|_{\text{op}} \leq B \sum_i \|T_i(w_i)\|_F. \quad (12)$$

The implemented Frobenius relaxation is a sum of Euclidean group norms. It ignores some sharing of δ across particles and can be conservative. Charging the entire first-order interaction to an isotropic remainder was substantially worse in development.

The same construction operates in the full supported voxel space. For voxel position x , box length d , rotation radius a and shift radius s_t , set

$$L(q, x) = \frac{2\pi}{d} (a \|k\| \|x\| + s_t \|q\|), \quad (13)$$

$$H(q, x) = \frac{2\pi}{d} a^2 \|k\| \|x\|. \quad (14)$$

The absolute second derivative of its complex Fourier column is bounded uniformly by $|C_i(q)|[L(q, x)^2 + H(q, x)]$ before whitening. Summing this column bound against $|\rho_0(x)|$ gives the pilot bound; its frequency/voxel Frobenius norm gives the operator bound. Analytic first derivatives supply the full-density interaction. Thus the implemented joint certificate covers the stated ambient density ball and local pose balls simultaneously. Fixed CTFs and known whitened Gaussian noise remain assumptions; CTF error, window correlations and heterogeneous densities are not covered by assertion.

5.1. Curvature and shared-density post-audits

Retaining the quadratic pose terms replaces the second-order remainder with a uniform cubic one (Appendix A.4). It adds structured pilot and density Hessian terms, and can substantially reduce width. The density component can be

tightened further: it is one shared error across particles, not one independent error per image. Let $h = \mathcal{A}^*w - \ell$ and collect blocks $H_0 = h$, $H_i = T_i(w)$ and $H_{n+i} = U_i(w)/2$, where U_i contains all ordered second-derivative adjoint columns. Their coefficients are $1, u_i, u_i \otimes u_i$, each of norm at most one. For any positive fixed block scales d_j ,

$$\left\| \sum_j H_j v_j \right\| \leq \sqrt{\sum_j d_j} \| [H_j / \sqrt{d_j}]_j \|_{\text{op}}. \quad (15)$$

Multiplication by B bounds the combined density residual and interactions. The pilot and cubic remainder terms remain unchanged. Equation (15) is an elementary operator-norm inequality. We compare it with separate spectral and triangle bounds, use $d_j = \|H_j\|_F$ for nonzero blocks, and retain the smallest valid bound. This post-audit uses fixed weights and preserves coverage; it does not inherit an optimality certificate for the new width.

First and second derivative adjoints require only five first and fifteen distinct second Fourier moment fields per particle. Direct summation or batched nonuniform transforms evaluate these fields on a larger supported grid without its pixel-space derivative Grams. Physical voxel-volume scaling, the inference center and the target all use that audit grid. This verifies larger finite density spaces, not arbitrary unresolved continuous density.

6. Development Experiments

All results below are exploratory. A confirmatory protocol has not been frozen. Experimental particle pools have already been inspected during the earlier reconstruction study and subsequent development; their partitions must not be described as untouched final tests.

6.1. Data, references and provenance

We retain 8,192 extracted experimental images from each of EMPIAR-10028, 10049 and 10076 (EMPIAR, 2015a;b; 2016). Source exposure groups are reconstructed from deposited metadata. EMPIAR-10076 requires the original Frealign FILM field because its STAR micrograph numbers are sequential particle identifiers. Hash-based whole-group partitions provide pilot, tuning, inference-half and test labels within this development pool. Consensus poses and CTFs remain conditional inputs.

Deposited maps EMD-2660, EMD-6487 and EMD-8434 define independent voxel signal generators after explicit low-pass resampling. They are simulation truth, not proof of experimental accuracy. Nonuniform Fourier transforms project these maps independently of the Gaussian representation. Native and downloaded-file hashes, source indices, signs, field sizes and software versions are recorded.

6.2. Representation and exact-dictionary controls

A noiseless projection audit with 256 particles per split and a radius-six frequency band found relative held-out errors of approximately 1.8%, 2.7% and 7.7% for the finest tested Gaussian dictionary. These residuals demonstrate why noise-only intervals do not cover representation error.

The bounded-model coverage benchmark has 48 settings: 12 seeds, broad versus preferred orientations, and rotation radii 0.5° or 2° . Each setting uses 96 particles, 40 coefficients, 20 random bounded signals and estimator-specific adversarial controls. We report exact Gaussian noise-marginal coverage and cross-check with 2,000 paired noise draws per signal. Shared designs and paired noise do not constitute independent voxel trials.

The exact-dictionary comparison is plotted in Appendix D. The structured certificate’s smallest tested coverage was 0.991 at nominal 0.95; omitting the quadratic remainder gave 0.932. Median width relative to the no-data bound was 0.557 for broad views at 0.5° , and approximately one for preferred views. This is a usefulness limitation, not a calibration success to conceal with pooled averages.

6.3. Ambient-space audit

The fixed-pose audit uses 128 real acquisition geometries per dataset, a 24^3 voxel grid, a spherical support of radius 0.35 times the field, and prescribed signal-to-noise power ratio 0.1. A pilot derived from real particles and each reference generator are normalized to unit voxel Euclidean norm. We set $B = 2$ before seeing coverage; the triangle inequality covers all such normalized generator/pilot pairs. This simulation convention does not estimate the density radius of experimental data.

The initial dictionary has 33 non-null Gaussian modes. Two targets, a central average and an axial contrast, use Gaussian averaging widths 0.03 and 0.07 times the field. Restricted intervals, the same weights with a full-grid audit, enriched weights, and the ambient matrix-free solution use identical estimands. The larger-grid model is deliberately able to express errors outside the initial dictionary.

Table 1 reports median widths across the two targets. The full-space solver reached its 0.5% objective-gap tolerance in 12 of 12 cases, taking 2.4–11.1 seconds per target on the Mac. The enrichment solver reached the same outer tolerance in 8 of 12 cases within 100 rounds; nonconverged results retain valid bounds but do not establish optimality. The solvers optimize the conservative sum-width, so their sharper reported widths need not be ordered exactly by their objective gaps.

Dictionary-only intervals can be narrow because they exclude admissible density errors. For the EMPIAR-10028

Table 1. Fixed-pose ambient audit: median interval width relative to the no-data width across two density targets. All three methods after auditing use the same supported voxel class. The first column instead assumes the initial 33-dimensional dictionary is correct and is not an honest interval for that larger class. All entries are development results.

Dataset	Averaging width / field	Dictionary only	Same weights, audited	Enriched	Full matrix-free
10028	0.03	0.008	0.978	0.689	0.687
10028	0.07	0.007	0.510	0.053	0.050
10049	0.03	0.020	1.016	0.718	0.716
10049	0.07	0.020	0.636	0.075	0.073
10076	0.03	0.024	0.949	0.721	0.719
10076	0.07	0.023	0.462	0.072	0.072

Table 2. Joint ambient/pose interval width divided by the no-data width. Columns 2–4 use a 100-iteration cap. The last column repeats 2° with up to 500 iterations; EMPIAR-10049 retains a 0.548% gap, above the 0.5% tolerance. Others meet tolerance. These are different optimization budgets, not separate independent replications.

EMPIAR	0.1°	0.5°	2°	2° , longer
10028	0.087	0.200	0.866	0.865
10049	0.118	0.239	0.996	0.992
10076	0.110	0.218	0.910	0.907

central broad average, analytic coverage on the independent voxel-map generator is approximately 4.2×10^{-5} . The same image weights, with the ambient bias bound, cover at the prescribed level over the ambient ball. On each estimator’s constructed ambient bias boundary, the audited intervals attain nominal 0.95 Gaussian coverage up to numerical tolerance. These boundary calculations check the theorem and implementation; they are not independent evidence of practical class calibration. Broader targets have substantially smaller relative widths than fine targets. The prescribed norm radius and fixed-pose assumption are essential to interpreting that result.

6.4. Joint ambient and nonlinear pose uncertainty

We combine the full supported voxel space with the nonlinear pose bound, using the broad axial contrast, 128 particles per geometry and the same prescribed density radius. Rotation radii are 0.1° , 0.5° and 2° , with a shift radius of 0.01 working pixels in their joint unit ball. Signal cases include random density/pose boundaries, estimator-directed adversaries and the independent voxel reference under a coherent rotation. Table 2 shows that a valid but conservative interval can still be uninformative at a large nuisance radius. Fixed-pose intervals reach coverage as low as 0.904 on these allowed perturbed cases. The nonlinear certificate’s smallest tested analytic coverage is 0.9997; its conservatism requires scrutiny rather than being interpreted as superior calibration.

The completed curvature comparison spans three geometries, two targets and three pose radii. All 18 quadratic solves reach the 0.5% objective-gap tolerance. At 0.5° , their width/no-data ratios are 0.128–0.178, versus 0.423–0.532 at 2° . Figure 1 shows the full comparison, including probability of establishing the correct target sign. Central averages become detectable at the intermediate radius; the deposited-map axial contrasts remain too small. Their averaging standard deviation is 0.07 times the field (16.5–33.8 Å), so this does not establish atomic-scale utility.

For both targets at 0.5° and 2° , the shared-density post-audit reduces the same weights’ interval widths to 0.50–0.65 of their original values. Independent local nonlinear stress tests optimize pose errors while eliminating the density ball exactly. Four starts for each bias sign find feasible adversaries, recomputed in float64 after MPS search; they supply lower bounds on worst bias, not global solutions. Their comparisons with the upper bounds are retained in the repository.

A finer-feature development probe uses a 32^3 grid, frequency radius 12 and averaging width 0.03 times the field. At 0.5° on EMPIAR-10028, the first-order bound equals the no-data interval; retaining curvature reduces its relative width to 0.449. The expanded comparison is still running and no multi-dataset high-bandwidth conclusion follows from this probe.

6.5. Bootstrap tradeoff and finite-grid stability

A fixed-pose benchmark fits 33- and 123-mode Gaussian models and all 2,517 supported voxels. It uses 999 source-group bootstrap resamples, four prescribed ridge strengths and 1,000 independent Gaussian measurement-noise replications per setting. Percentile/basic intervals, sampling covariance, posterior covariance and a same-estimator ambient bias audit are recorded. The group-resampling plan is reused across noise draws; binomial error bars are conditional on that plan.

Weakly regularized full-voxel bootstrap intervals have center coverage 0.938, 0.956 and 0.958 across the three geometries,

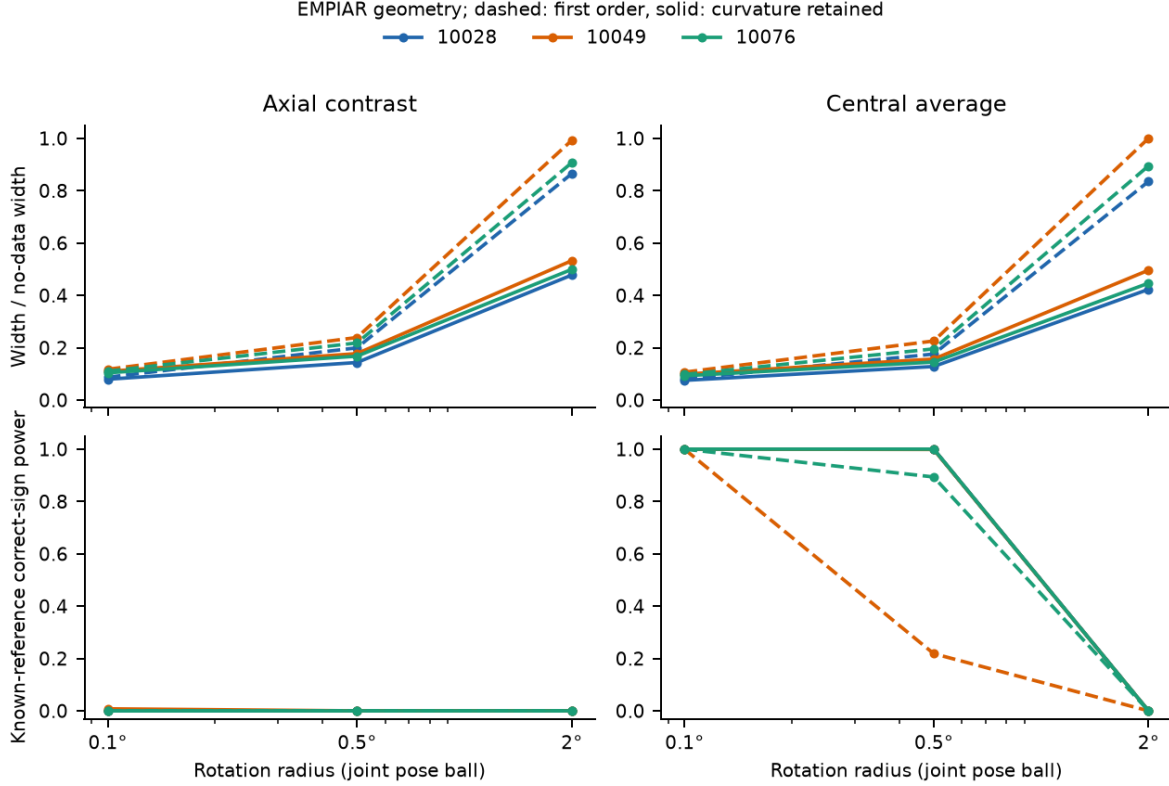


Figure 1. Curvature retained versus a first-order expansion with bounded second-order remainder, under the same density and pose classes. Dashed curves use the first-order procedure; solid curves retain curvature. The lower panels use the independent deposited-map generator under a coherent allowed rotation. Sign power and interval width expose limitations that high coverage alone conceals. These intervals precede the shared-density post-audit.

with widths smaller than the robust audit. Strong regularization can cause substantial undercoverage: EMPIAR-10028 center coverage drops to approximately 0.338. A conservative bias audit therefore does not dominate bootstrap width in every setting. Figure 3 retains the regularization sweep, including cases favorable to bootstrap.

A fixed-pose refinement study audits 48 coarse-grid solutions on a common 64^3 supported space. Grids 16, 24, 32 and 48 preserve physical field, density-energy norm and noise units. Width ratios between fine audit and original calculation range from 0.72–1.45 at grid 16 and approximately 0.999–1.006 at grid 32. Nonlinear Fourier-moment audits additionally cover both targets, both pose radii and all three geometries on grids 24, 32 and 64. Discretizations are not identical nested classes; ratios need not increase monotonically. Agreement is a numerical sensitivity check, not a continuous-density guarantee.

6.6. Experimental reconstruction controls

We execute stock cryoDRGN 4.3.1 homogeneous neural training on both exposure-group halves of all three stacks:

three hidden layers, width 256, batch size eight, fixed consensus poses and CTFs. These are actual neural fits; they are not ab initio inference or uncertainty estimates. Gaussian and voxel ridge fits use the same halves. All methods are evaluated on identical held-out particles and 1,410 Fourier pairs per image inside radius 30 of the 64-pixel grid, with the same window and image-background normalization. Neural training itself uses the stock radius-32 loss and per-half global scaling, so training objectives are not identical.

Table 3 selects checkpoints using tuning-group error only, from runs through 60 epochs for EMPIAR-10028/10049 and 100 epochs for EMPIAR-10076. The selected epochs are 60, 50 and 100. The development plateau check motivated these budgets; it is not a preregistered stopping rule or a proof of global convergence. Exposure-group bootstrap comparisons, checkpoint hashes and direct-network versus map-Fourier checks are recorded with the results. FSC assesses conditional map agreement; neither it nor image prediction labels true experimental density coverage.

Table 3. Experimental-image prediction error on development test groups (normalized MSE; lower is better). Neural checkpoints minimize tuning-group error among completed runs: epochs 60, 50 and 100 respectively. All half-map FSCs remain above 0.143 through the sampled limit in the last column, which is not a measured crossing.

EMPIAR	Gaussian	Voxel	Neural	Limit (Å)
10028	0.8355	0.8417	0.8293	16.08
10049	0.9308	0.9316	0.9302	7.87
10076	0.8764	0.8806	0.8707	13.97

6.7. Baselines and numerical checks

Implemented matched mathematical baselines include full Gaussian posterior covariance, diagonal variational covariance, regularized sampling covariance, joint Gaussian pose covariance, identifiable unregularized estimation, fixed-pose bias-aware intervals, coarse and structured nuisance bounds, and the no-data interval. These are not executions of every external paper’s original code. Full Gaussian posteriors pass their exactly matched linear prior-predictive identity at 0.95. Narrow priors can nevertheless fail bounded fixed-signal tests; wider priors and unregularized estimation can produce very wide intervals.

Independent conic optimization checks the majorization, primal-dual and ambient solvers on small instances. Derivative finite differences, nonlinear remainder tests, direct Fourier sums, adjoint identities and independent Gaussian-integral checks validate implementation components. The current suite has 38 passing tests, including a three-dimensional Fourier-transform check for externally generated maps. Tight FINUFFT tolerances are numerical checks, not interval-arithmetic guarantees on roundoff.

7. Limitations and Work Remaining

The current evidence does not establish an end-to-end uncertainty method with estimated poses, noise law and density class. Local pose radii and the total density bound are assumptions. Selecting them from the same inference residuals would require new analysis; a narrow residual does not rule out an almost invisible density error. Learned support can exclude true structure. Gauge and hand ambiguity must be fixed or included in the target class.

The next validation must freeze the method before fresh simulations and additional source groups, extend higher-bandwidth and independent-generator comparisons, and test correlated noise, CTF errors, heterogeneity and support violations. Bootstrap, curvature, grid-sensitivity and neural-checkpoint development comparisons are now available. The power analysis above is only an initial coarse-feature check. Simultaneous maps need multiplicity correction.

Real-data prediction and stability cannot supply unavailable density-coverage labels. Comprehensive literature screening and independent review of a complete candidate remain outstanding.

Reproducibility and Impact

The active research branch is `uncertainty-study` at <https://github.com/yashpatel5400/fourier-cryo-splats>. It preserves failed development approaches and distinguishes them from completed outputs. The earlier reconstruction release v0.1.0 remains available with its conditional FSC results; it is not the uncertainty study’s accuracy claim. No new conference submission or acceptance is implied by this format. Background PDFs are retained locally with source hashes rather than redistributed. Prematurely calibrated-looking maps can mislead structural interpretation; explicit assumptions and negative controls are central outputs of this work.

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A. Additional Derivations

A.1. An identifiability lower bound

Proposition 1. *Let two allowed parameters induce $N(\mu_0, I)$ and $N(\mu_1, I)$, with targets t_0, t_1 . Any interval covering each target with probability at least $1 - \alpha$ has expected length under the first law at least*

$$|t_1 - t_0| [1 - 2\alpha - \text{TV}(P_0, P_1)]_+, \quad (16)$$

where $\text{TV}(P_0, P_1) = 2\Phi(\|\mu_1 - \mu_0\|/2) - 1$.

Proof. Under P_0 , the probability of containing t_1 is at least $1 - \alpha - \text{TV}(P_0, P_1)$ by the definition of total variation. The union bound gives simultaneous containment of both targets with probability at least the displayed positive part. The length on that event is at least their separation. The total variation formula follows by projecting onto the line between the Gaussian means. $\square$

This standard two-point argument also applies to different allowed nuisance settings. Exact null directions give a nonvanishing uncertainty floor even with perfect image prediction. It does not assert that a finite numerical search finds the largest possible separation.

A.2. Gaussian density-energy Gram matrix

For isotropic paired features, let

$$P_{jk} = \left(\frac{2\pi\sigma_j^2\sigma_k^2}{\sigma_j^2 + \sigma_k^2} \right)^{3/2} \exp \left[-\frac{\|\mu_j - \mu_k\|^2}{2(\sigma_j^2 + \sigma_k^2)} \right], \quad (17)$$

$$Q_{jk} = P_{jk} \text{ with } \mu_j - \mu_k \text{ replaced by } \mu_j + \mu_k.$$

Integrating products of Gaussians and applying Parseval gives real and imaginary coefficient blocks $2(P + Q)$ and $2(P - Q)$, with zero cross block. A zero-centered imaginary coefficient is exactly null. Numerically excluded eigenmodes are reported; exclusion changes the stated Gaussian subspace. In an ambient audit they remain available to the larger space rather than being silently assigned zero uncertainty.

A.3. Uniform nonlinear remainder

For a Gaussian of width σ at radial distance r , bounds on value, gradient and Hessian are

$$g_0(r) = e^{-r^2/(2\sigma^2)}, \quad (18)$$

$$g_1(r) = r g_0(r) / \sigma^2, \quad (19)$$

$$g_2(r) = (1 + r^2/\sigma^2) g_0(r) / \sigma^2. \quad (20)$$

On an interval of possible distances, g_0 is largest at the lower endpoint; g_1 and g_2 attain their largest value at the point

nearest σ . A rotation radius $a \leq \pi$ gives displacement at most $2\|k\| \sin(a/2)$, speed at most $a\|k\|$ and acceleration at most $a^2\|k\|$. A shift radius s gives phase speed at most $2\pi\|q\|s/\text{box}$.

The product rule bounds the second derivative by the sum of squared-speed times g_2 , acceleration times g_1 , twice speed times phase-speed times g_1 , and squared phase-speed times g_0 . Sum conjugate-pair bounds and apply the fixed CTF/noise scale. A coefficient-column Frobenius bound controls the operator term; the pilot-weighted bound controls its pilot term. Taylor's integral formula gives Equation (11). Evaluating a Hessian only at the nominal pose would not establish this uniform result.

A.4. Retaining quadratic pose terms

The isotropic quadratic remainder can dominate an interval. A development extension retains the curvature and uses a cubic remainder instead. Put $E_{iab} = \partial^2 A_i / \partial u_{ia} \partial u_{ib}$ at zero, $Q_i(w)_{ab} = w_i^\top E_{iab} \rho_0$, and let $U_i(w)$ collect columns $E_{iab}^\top w_i$ over all ordered pairs. Since $\|u_i \otimes u_i\|_F \leq 1$, a valid bias bound is

$$\begin{aligned} b_2(w) = & B \|\ell - \mathcal{A}^* w\| + \sum_i \|J_i^\top w_i\| \\ & + B \sum_i \|T_i(w)\|_F + \frac{1}{2} \sum_i \|Q_i(w)\|_F \\ & + \frac{B}{2} \sum_i \|U_i(w)\|_F + \sum_i \gamma_{3i} \|w_i\|. \end{aligned} \quad (21)$$

For a Fourier voxel column, define $T(q, x) = 2\pi a^3 \|k\| \|x\|/d$. Its third derivative along any allowed pose segment is bounded by $|C|[L^3 + 3LH + T]$ before whitening. The three summands follow by differentiating the phase exponential three times. Pilot contraction and a column Frobenius bound give H_{3i} and K_{3i} , hence $\gamma_{3i} = (H_{3i} + BK_{3i})/6$. Taylor's integral remainder and Cauchy-Schwarz then give Theorem 1 with b_2 . Sharing of density errors and rank-one quadratic pose constraints are still relaxed.

Analytic column Hessians and pixel-space Gram factors implement these Euclidean groups. Mixed finite differences and independent nonlinear projections test the implementation. The full three-geometry, two-target comparison is shown in Figure 1. The spectral post-audit follows Equation (15); its proof is the operator-norm bound applied to the concatenated vectors $[\sqrt{d_j} v_j]_j$, whose squared norm is at most $\sum_j d_j$.

A.5. An independent noise-scale upper bound

Known noise is not necessary if an independently calibrated upper bound is available. A standard construction assumes independent calibration coordinates $z_j = \mu_j + \sigma e_j$,

$e_j \sim N(0, 1)$, with arbitrary fixed means and independent inference noise. For ν coordinates, set

$$\bar{\sigma} = \|z\| / \sqrt{\chi_{\nu, \beta}^2}. \quad (22)$$

The probability that $\bar{\sigma} \geq \sigma$ is at least $1 - \beta$: the central chi-square pivot is exact at zero mean, and a noncentral chi-square has a smaller lower tail. Conditional on calibration, weights may use $\bar{\sigma}$. On this event their inference standard deviation is at most $s_+ = \bar{\sigma} \|w\|$. For $\alpha < 1/2$, $q(s_+, b, \alpha) \geq b$, and Gaussian coverage decreases with the actual noise standard deviation up to s_+ . Integrating the conditional bound yields coverage at least $(1 - \alpha)(1 - \beta) \geq 1 - \alpha - \beta$.

This extension does not justify independence of windowed Fourier pixels or estimate arbitrary covariance. With 128 calibration coordinates and $\beta = 0.01$, a 20,000-draw development control gives scale-bound failure 0.0103 under independence. Equicorrelation 0.5 raises that failure to 0.503. Such a calibration set violates the premise even if a particular resulting interval remains conservative. Experimental whitening remains unresolved.

A.6. Simultaneous targets and sensitivity budgets

For K targets and parameter-budget choices fixed independently of inference noise, applying Theorem 1 with α/K gives simultaneous coverage by the union bound. Correlated Gaussian maxima can reduce the noise component if the joint weight covariance is accounted for. Selecting the most favorable point on a sensitivity surface after inspecting inference data requires such simultaneous control. A density radius supplied by simulation truth is an oracle control and cannot be reported as a practical calibration method.

B. Survey and Experimental Audit Trail

The repository’s living survey distinguishes density uncertainty, pose uncertainty, conformational variability/populations, map-processing confidence and atomic-model uncertainty. Its broad Europe PMC retrieval ledger contains 16,969 deduplicated records, mostly application papers; that count is not a count of reviewed methods. The curated list contains 89 candidate works at this draft checkpoint, with local full-text availability recorded separately. Recent items include LocScale-2.0, CryoDiff, CryoSplat, GEM, measurement-limited heterogeneity, population-counting work and the CAHRA dataset-construction preprint. Preprints and final publications are distinguished, and titles/versions are checked against primary sources.

The full experimental protocol draft records unimplemented comparisons and required failure controls. Fixed-pose resampling must be distinguished from rerunning alignment;

a mathematical diagonal-Gaussian control must be distinguished from reproducing Ullrich et al.’s full software; cryoDRGN backprojection must be distinguished from training a neural model. Exposure-group splits do not undo the dependence introduced by full-data consensus poses. All development results, including the initially incomplete covariance-factor benchmark, remain identifiable by filename and status.

C. Uncertainty Targets in the Recent Literature

The following distinctions guide baseline selection. They do not rank methods that solve different problems on one inappropriate score.

Pose inference versus pose confidence. Posterior-averaged orientation estimation (Xu et al., 2026) and probabilistic orientation search in SIMPLE (Van et al., 2025) improve pose estimation through different statistical or optimization constructions. Neither an orientation distribution nor a stochastic search neighborhood should be relabeled a calibrated per-particle confidence ball without verifying its guarantee. Our prescribed local balls avoid that claim, but consequently do not solve practical pose calibration.

Physical heterogeneity versus estimation error. RECOVAR estimates covariance and uses kernel regression to account for noisy latent coordinates (Gilles & Singer, 2025). DynaMight and 3DFlex represent motion (Schwab et al., 2024; Punjani & Fleet, 2023). Particle-counting work explicitly studies when assignments misestimate state populations (Evans et al., 2026). These motivate separate tests of population recovery, structural variation and uncertainty in a homogeneous density. The present bounded-density target covers only the last. CryoDRGN-AI addresses joint ab initio reconstruction (Levy et al., 2025); fixed-pose homogeneous training does not reproduce that task.

Map validation and learned priors. False-discovery confidence maps test density significance against a noise reference (Beckers et al., 2019). Directional resolution measures acquisition anisotropy (Vilas et al., 2020); independent-particle tests assess map fit (Ortiz et al., 2020). Bias/variance analyses explain why validation requires more than internal agreement (Sorzano et al., 2022). Blush introduces learned reconstruction regularization (Kimanian et al., 2024), while LocScale-2.0 attaches confidence to enhancement relative to its stated proxy (Bharadwaj et al., 2026). Matched evaluations must retain these targets. A sharper-looking map or a significant positive voxel is not itself a bound on error relative to unknown density.

Atomic ensembles and independent modalities.

Experiment-guided AlphaFold3 conditions structural generation on reconstructed ESP maps and other measurements (Maddipatla et al., 2026). Its atomic-ensemble target differs from raw-particle density inference. The multimodal examples also illustrate why improved map agreement need not imply improved fit to a separate measurement modality. Independent evidence is relevant to assessing structural interpretations that the density measurements alone leave ambiguous.

Simultaneous inverse inference. Constrained test-inversion methods already address joint confidence regions for multiple functionals (Batlle et al., 2025). A single data-consistency region can support simultaneous projections, whereas separately calibrated marginal intervals generally require an appropriate correction. Our present scalar-target theorem is not a new general framework for simultaneous inference, and a colored map of marginal intervals must not be presented as simultaneous coverage.

D. Additional Development Comparisons

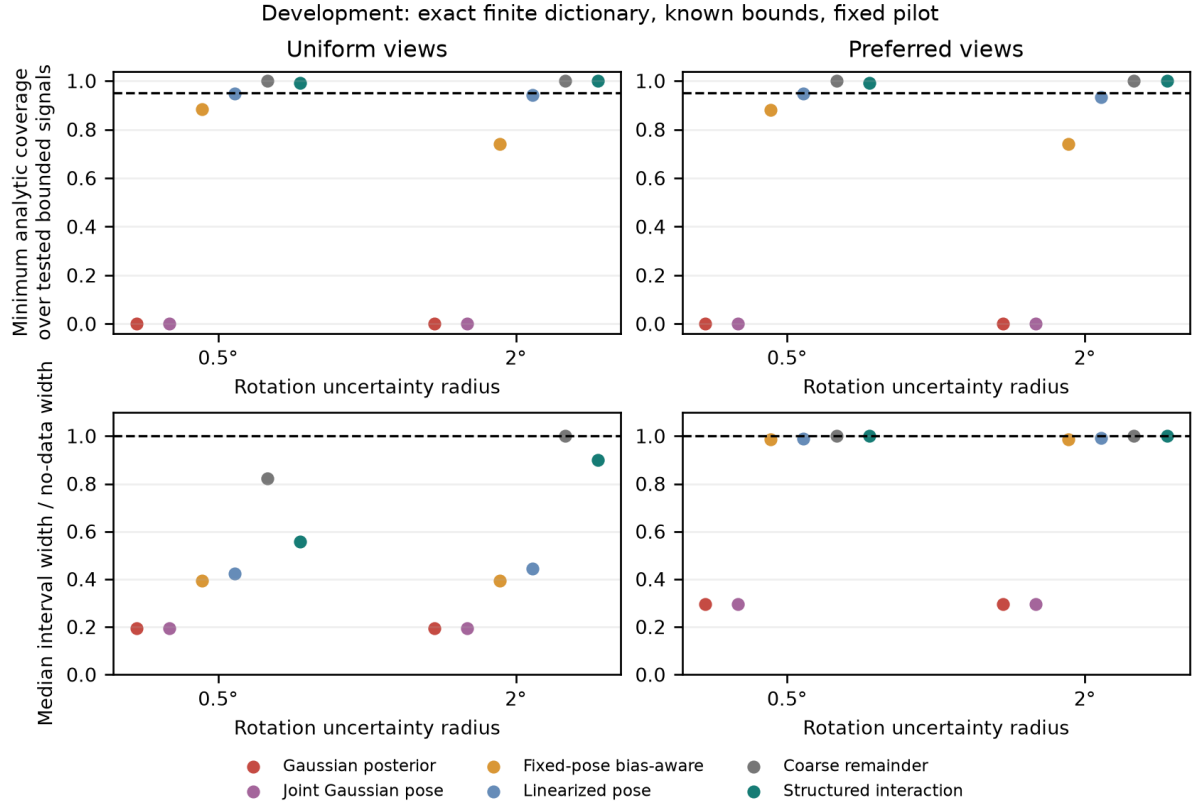


Figure 2. Exact-dictionary development tests with known bounds and an independent fixed pilot. Minimum coverage is over the finite tested set, including adversarial signals; it is not an empirical proof of the worst case. Widths count each independent design once. Gaussian posterior methods separately pass their correctly matched prior-predictive coverage check at 0.95. Their fixed-parameter boundary failures do not contradict Bayesian prior-average calibration.

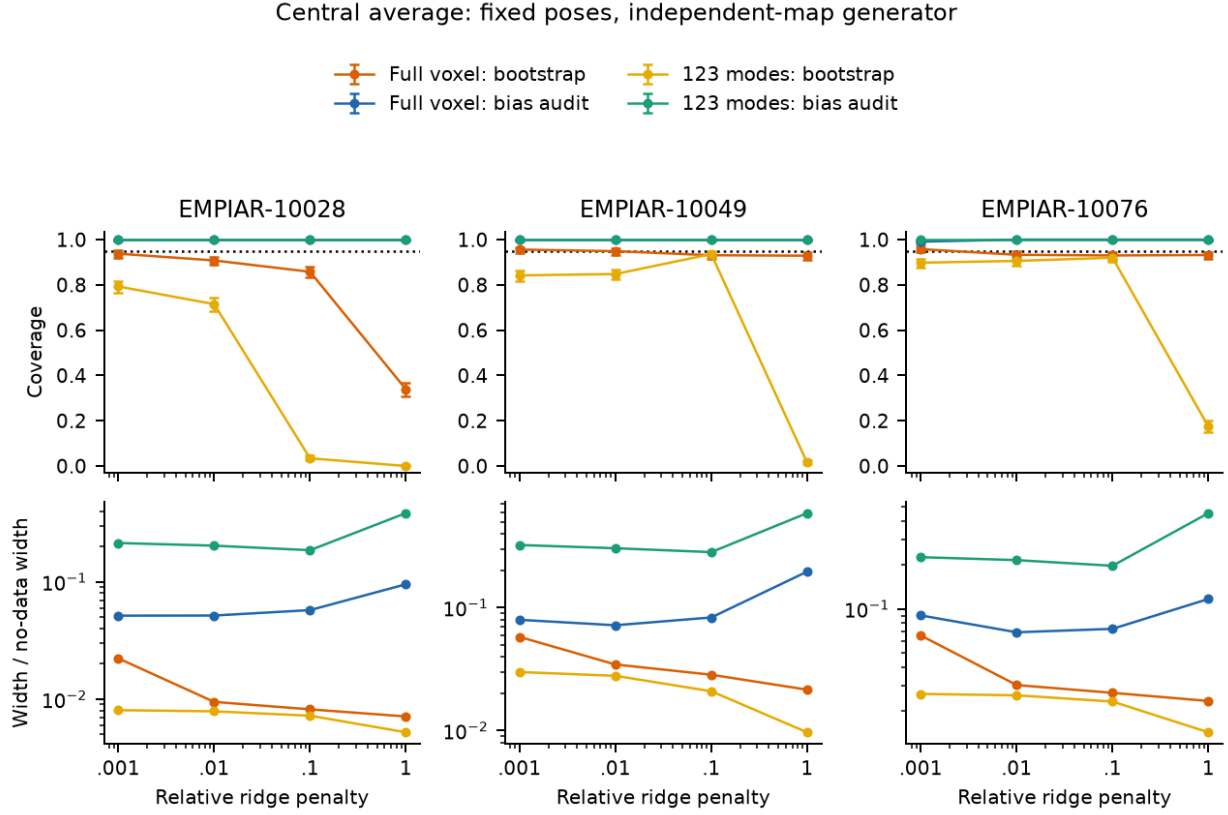


Figure 3. Central-average coverage and widths across all four ridge penalties. Each geometry uses its own fixed independent deposited-map generator. Error bars are binomial 95% intervals over independent measurement-noise draws, conditional on the reused group-bootstrap plan. Weak-regularization bootstrap can be narrow and near nominal; stronger regularization exposes bias. These tests do not establish empirical coverage for unknown experimental density.

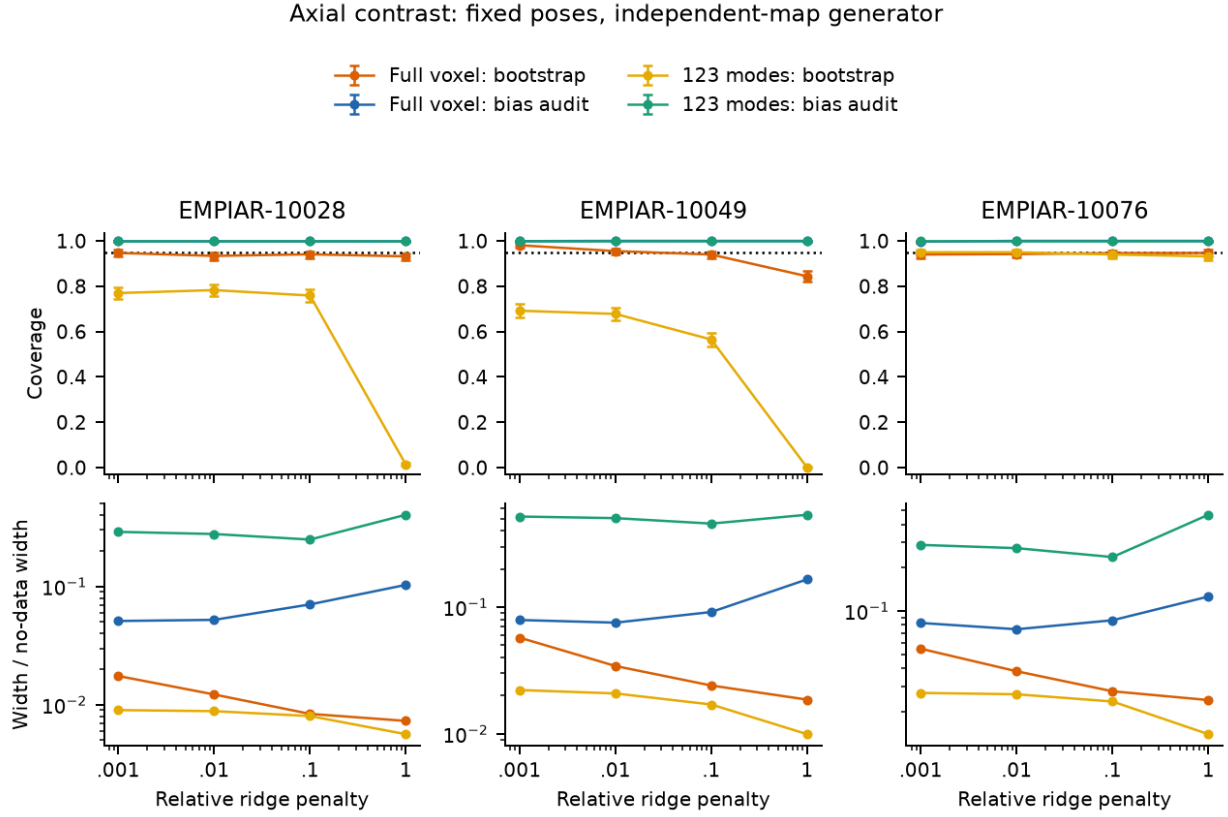


Figure 4. The same complete regularization comparison for the axial contrast. Both the estimand and the unknown simulation signal are fixed before noise replication. The full results also retain the smaller 33-mode dictionary, basic bootstrap and Gaussian covariance controls.

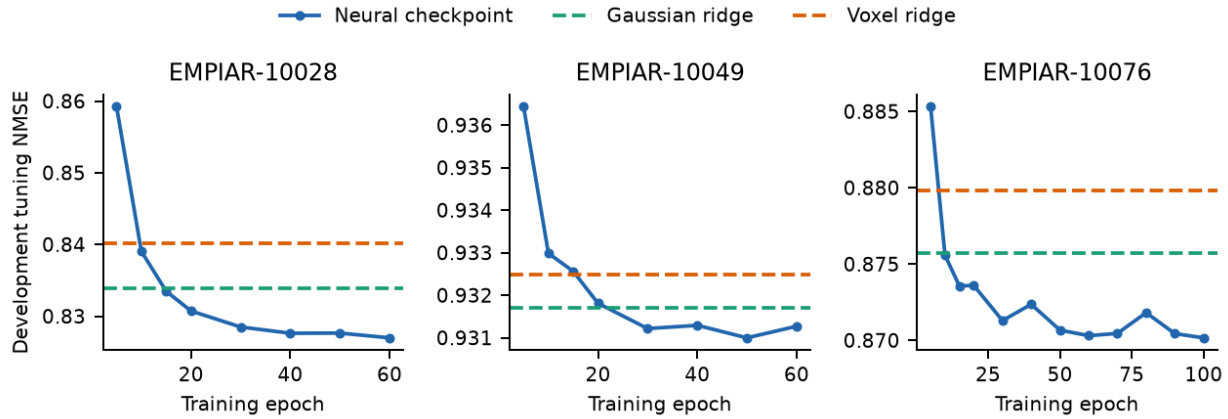


Figure 5. Development tuning error at all evaluated neural checkpoints, with the classical reconstruction controls. Checkpoint selection minimizes this tuning error; the test-group error is not used to select the checkpoint. Different training objectives and supplied consensus poses limit this comparison.